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PrepSmartly: Multimodal AI-Powered Interview Simulation System with Real Time Feedback and Adaptive LLM Evaluation

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***Abstract*— In the competitive job market of today, prepping for interviews involves more than just developing relevant technical skills; interviewers would like to see you able to communicate well and show confidence as well as give the right answers to behaviour-based questions. Unfortunately, most of the traditional resources available for preparing for job interviews use static question banks with little or no feedback delay. This does not reflect the nature of job interviews in the real world; many times, your interviewer will change their question based on your answers. As a solution to this issue, we present PrepSmartly, an experiential interviewing simulation with an AI-based multimodal framework, designed to simulate and create a more realistic practice experience. We achieve this by using speech analysis, facial recognition software, and code evaluation to produce a simulated interview environment. The platform uses voice processing technology, computer vision models, and two LLM (Large Language Model) based systems to create adaptive responsive questions and provide real-time evaluation. Another benefit of the solution is that candidates will be able to show their programming skills during technical interviews using an integrated online code editor. This system gives real-time micro-level feedback while the interview is occurring and provides candidates with a detailed performance report at the end that uses rubrics to assess various aspects of their interview performance such as: Technical Correctness, Clarity Communication Skills, Emotional Expression, Non-Verbal Cues.**

***Keywords*— Multimodal AI, Interview Simulation, Large Language Models, Speech-to-Text, Facial Expression Analysis, Code Evaluation, Real-Time Feedback, Adaptive Learning**

I. INTRODUCTION

Interviews are still an important part of the recruitment process, be it for a job, school or a future career opportunity. They are intended to measure a candidate's level of knowledge, as well as the ability to communicate that knowledge clearly and concisely. A candidate's confidence, ability to think clearly, ability to think through problems and how they respond behaviourally all contribute to how well they perform in an interview. While most candidates have prepared thoroughly for their interview, many of them do poorly because they are unable to communicate their ideas clearly during the interview or are nervous when they meet the interviewer.Interviews remain one of the most important steps in selection processes, whether for jobs, higher studies, or career growth opportunities. The purpose of these assessments is two-fold; to evaluate a candidate’s knowledge, and also to evaluate the means in which the candidate communicates the knowledge that they possess. Confidence, clarity of thought, ability to problem-solve, and overall behavior will greatly affect how well someone performs during the entirety of the assessment.. Even well-prepared candidates sometimes underperform simply because they struggle to present their ideas clearly or manage nervousness during the interaction.

Many professionals and students still do not have access to adequate training and resources for interviews even though they are critical components of converting your effort to a successful outcome. Many experienced mentors and coaches charge a lot of money for their time (having trained a person) and therefore those who do not have the budget to hire a mentor/coach find themselves at a disadvantage if they do not have access to a structured coaching program. Therefore, most people who prepare for interviews do so through self-practice or through free resources available on the internet, however, these resources may not adequately address the specific weaknesses of an individual candidate.

Most of the current online interview preparation solutions are primarily focused on the practice of answering "canned" or pre-defined questions and answers. Practicing these questions and answers can help with the review of information but does not accurately simulate the reality of participating in a live/in-person interview. The feedback you receive is usually very general and administered a week or more after completion of the interview preparation activity so you may not get a true picture of your individuality related to your communication skills and/or degree of self confidence. In addition, most of these tools only evaluate your written or typed responses and not the other areas of your performance such as your manner of speech, expressions on your face and/or any coding required in response to a live interviewer all together. Due to this

limitation, interviewees develop only an allegedly incomplete picture of how well prepared they are for a live interview and, therefore, may not possess the awareness of which areas of improvement are most critical for their success.

This paper introduces PrepSmartly, a multimodal AI-assisted mock interview platform which thus aims to resolve these limitations by creating a more realistic and responsive practice environment for interviewing. By utilizing advancements in speech analysis, computer vision, and large language model technologies, PrepSmartly can provide simulated interviews that change based upon an interviewee's performance.

II. LITERATURE REVIEW

In reference [1], the authors discuss an AI driven mock interview simulator which utilizes Generative AI (Gemini) and combines both video and analytics based on sentiment data in order to conduct a multi-modal (audio + visual) needs based analysis to produce appropriate domain specific and personalized questions for each user. Thus allowing users to gain realistic practice experiences during their simulated interaction with their computer system, as well as receive holistic feedback for both verbal and non-verbal communication performance. This method of creating adequate interview preparedness through an analysis of communication, tone and body language in a simultaneous manner will improve the user's level of confidence and overall preparedness. However, the fact that this model has a high degree of dependence on cloud-based AI technologies and requires a significant amount of computer processing/analysis resources creates challenges created by both potential accessibility limitations and inadequate cost-effectiveness for scaling appropriately.

In [2] scientists research multimodal AI architectures used for fusing together both input (imaging) data and clinical metadata by means of latest architectural developments (transformers, GNN's, and more), with the objective of improving the quality of decision making, through providing greater contextual awareness by fusing visual data with textual data. Therefore they show how multimodal systems can be used to increase accuracy of decision making through fusing together two different data modalities into one decision-making process. There are many potential uses of an integrated framework, such as combined methods for solving complex problems that could only be solved using multiple different methods; despite this the generalization capability of many multimodal AI systems has been significantly affected, due to issues associated with data sparsity (insufficient amounts of training data), data/image bias (to one type of image), and taxonomical inconsistency (different data/image classifications or categories).

Reference [3] describes a functional multimodal system for facial expression and sound analysis of interviews functioning as a virtual recruiter. It encompasses facial emotion recognition ability (via facial expression detection technology), voice recognition ability (via voice pattern analysis technology), Speech-to-text (STT) capability ability to assess User's performance in a given task. This multimodal system provides candidates with timely and objective feedback based on their current performance with regards to facial expression, clarity when speaking, and how quickly they respond to questions, providing candidates with an ongoing basis to keep track of their improvement throughout multiple sessions. However, the prototype only tests in laboratory conditions, limiting the system's target user population and ability to accurately assess performance in real-world contexts, thereby severely restricting scalability potential.

The evolution of facial expression recognition (FER) technologies is examined in-depth by a comprehensive survey examining both the advancement from traditional feature-based methods to today’s deep learning-based methods such as CNNs, RNNs, GANs, and Transformers[4]. More than 60 FER Models and 21 FER data sets were analyzed as part of this study providing a thorough taxonomic and benchmarking assessment of all current FER methodologies. Although there have been numerous advancements in FER and the field has continued to show great promise in the areas of human-computer interaction and affective computing, the limitations of dataset bias, overfitting, reduced robustness of models in uncontrolled environments continue to plague researchers in the field.

[5] outlines an AI chatbot that functions as a job interviewer. The chatbot parses resumes and provides interview questions based on the applicant's qualifications. The advantage of this method is that it allows for personalization of the questions and also makes them more relevant. However, the chatbot's ability to handle non-traditional resumes and/or irregular profiles may lead to a misalignment of context during the interaction.

According to source number [6], the framework covers a New-Generation Artificial Intelligence Interview simulated performance assessment. This assessment involves the training of consistent and up to date content using an extensive database of questions and answers gathered from various sources on the web. Although this provides adequate content to conduct interviews with students, the legality and ethics of data scraping could raise issues regarding the reliability of the model itself due to the use of data collected from unverified sources, which are known to contain inaccuracies.

Article [7] reviews FER approaches using SLL, MLL, and LDL methods; reviews machine learning and deep learning approaches; and reviews topics related to these fields including healthcare-related applications, surveillance-related applications, human-computer interaction-related applications. There are many persistent challenges in each of the aforementioned categories including occlusion, pose variations, age differences, and lighting inconsistencies.

The authors of the article in [8] designed an artificial intelligence powered mock interview evaluator that uses a convolutional neural network to evaluate emotion, confidence, and knowledge of the subject using facial expressions, gestures, vocal tone, and vocal clarity determined from audio/visual data. Using this data, the evaluators provide segmented analysis of performance. However, the possible amount of data could become overwhelming; and the content does not provide as nuanced an interpretation as when being evaluated by a competent human evaluator.

In [9], subjective answer evaluation using natural language processing and machine learning is researched based on more effective slicing, merging, and converting of audio sources into one another; it performs well for pre-processing data only, as it does not have much of an ability to analyze audio beyond basic processes and does not help with analyzing audio based upon deeper meanings nor emotional meanings of the audio.

The investigation described in [10], provides a comprehensive analysis of natural language processing (NLP) methods employed for spontaneous speech-to-spontaneous speech summarization. Although the methods create higher levels of clarity within the summarization process and improve relevance determinations for each summarization generated, they rely heavily on the accuracy of the recognition process and do not easily capture the spontaneous nature of human speech.

Current research illustrates how Artificial Intelligence could support you to adequately prepare for interviews; however, nearly all available systems are limited to the use of either only a very partial multi-modal analysis or merely an array of static evaluations.

There are a lack of total integrated systems that offer adaptive questioning, real-time feedback, multiple modes of behaviors being assessed, and coding evaluation.

This study proposes a new system, "PrepSmartly," which puts together all of the things just listed to provide a single adaptive multi-modal simulation framework for interview preparation.

III. PROBLEM STATEMENT

Preparation tools are essential for candidates to be successful in today's highly competitive job market, especially those which can assess not only candidates' technical knowledge but also their communication abilities, level of confidence, and other common behavioral traits. However, existing tools for preparing for interviews generally use only fixed question banks and scripted question/answer sequences; thus, they provide mostly low-quality, generic feedback after each interaction; and too many of these systems do not provide enough realism. Non-verbal feedback, such as body language, facial expressions, tone of voice, and gestures, are critically important to the interview process; however, this type of feedback generally does not occur within existing preparation platforms. Further, concerns about the secure handling of audio and video recordings have been raised with many current platforms, leading candidates to be hesitant to use them.

Therefore, the proposed Multimodal AI-Powered Interview Simulation System with Real-Time Feedback and an Adaptive LLM Evaluation Metric (PrepSmartly), aims to provide candidates with an adaptive, privacy aware interview preparation solution that will enable them to practice all aspects of the interview process before they actually go on a real interview. The PrepSmartly system combines facial and body posture analysis through MediaPipe's Holistic Model and speaker and emotion analysis through DeepGram's Speech-to-Text APIs. In addition, there is a built-in code editor that utilizes the Piston model for practical coding assessment.

The PrepSmartly platform utilizes a dual Language Model-based (LLM) methodology, with one LLM that is responsible for generating adaptive interview questions, and another that will evaluate the candidate's response in accordance with established standards for accuracy, clarity, and relevance. The candidate will receive comprehensive verbal and non-verbal performance feedback through real-time analytics; and detailed post-session reports will be provided. As a result of the PrepSmartly platform, candidates will have the opportunity to develop or improve their technical competencies in addition to developing or improving their communication and behavioral competencies during their respective interviews.

IV. OBJECTIVES

1. To develop a voice to first type, adaptive impeachment simulation system that creates real impeachments using voices as the main way people communicate and interact with each other.
2. To create a real-time micro feedback system that will can find other people's problems while they are doing their interview; for example, they have problems such as lack of confidence, inappropriate rate of speech, inappropriate tone of voice, and lack of interaction with others.
3. To generate an evaluation model that uses multiple methods including but not limited to: vocal tone of voice, clarity of speech, facial expressions, findings of confidence and quality of responses, when reviewing employees' performance.
4. To produce a report after each interview session, providing a detailed summary report (structured) containing both positive and negative feedback with specific examples of how to improve their performance.
5. To provide other interview types besides the traditional full mock interview ie; self-paced guided interview with real time coaching, and interview question drills based upon rapid-fire topic selection by participant.

V. Proposed Solution

PrepSmartly is an AI-powered interview simulation platform that uses multiple modes (audio, video, text) to be a real-world alternative to traditional mock interview products. The platform bridges the gap between traditional theory-based interview preparation tools and actual interview performance by using the latest Large Language Model (LLM), speech, facial cue detection, and learning analytics technologies.

By having a dual LLM architecture with multiple modes of input, PrepSmartly can generate questions intelligently and control their level of difficulty adaptively, as well as evaluate a user's overall performance holistically. PrepSmartly has three different modes of conducting an interview: simulated, guided practice, and drill; and supports all five different types of interviews: HR, technical, coding, systems design, and domain-specific. Therefore, PrepSmartly meets the diverse needs for preparing for interviews.

There are four main goals for the proposed solution:
1. To recreate a realistic interview environment using voice-first interaction and a human-like interviewer.
2. To create individualized and adaptive questions based on how well someone has performed in a past interview.
3. To measure all possible aspects of someone's performance through multimodal assessment including speech, facial-cue based and text-based responses.
4. To give structured, actionable feedback and long-term progress analytics based on what they did during their interviews.

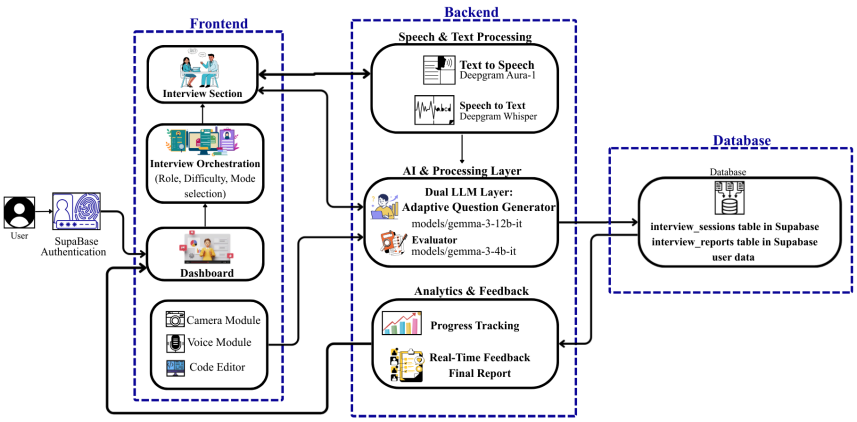

fig. 1- Proposed System

Fig. 1 presents the overall architecture of the proposed system, which is composed of three major layers:

1. Frontend Layer
2. Backend Layer
3. Secure Data Storage Layer

These layers communicate through RESTAPIs and event driven messaging. This approach ensures the system remains modular, scalable.

The architecture follows a microservice-oriented design. Each core function such as speech processing, large language model inference, analytics, authentication, and orchestration operates as an independent service. Although these services function separately, they work together seamlessly to deliver a better user experience.

A. Frontend Components

1. Interview Interface :

The Interview Interface acts as the primary point of interaction between the user and the virtual interviewer. It supports both voice-based and text-based communication, allowing users to respond in the format they are most comfortable with. The interface displays interview questions, captures user responses, and provides real time feedback when Guided Mode is enabled. For time-sensitive questions, built-in countdown timers help users manage their responses effectively and stay within the allotted time.

To further enhance clarity and confidence, the system includes a live transcription feature. As users speak the words are converted into text and displayed instantly on the screen. This allows them to verify that their responses are being captured accurately.

Overall, the Interview Interface is designed to create a realistic, engaging, and reliable interview experience while keeping the user fully informed and in control throughout the session.

2. Interview Orchestration Module :

The system provides for the Interview Orchestration Module to be configured by the user prior to the start of an interview session to provide a personalised and structured interview experience. The user can select their preferred type of interview—HR, technical or coding—based on what they require to prepare. The user can then set the level of difficulty of the interview to match their current skill and preparation level by choosing a level of Easy, Medium or Hard. The system also allows the user to practice in different modes including simulated, guided and drill practicals—based upon the users own way of learning—and choose the role/dimension of the individual (i.e. Software Engineer, Cloud Engineer etc.) to obtain role-specific questions. The parameters selected by the user will be sent to the backend and used for the generation of questions as well as evaluation processes for the overall interview session to match the configuration set by the user.

3. Dashboard:

The user dashboard offers an easy-to-read format of the user's performance and historic interview data for effective self-evaluation and tracking of performance over time. The user can also see a summary of the interviews they have completed and how consistent or inconsistent they have performed on each interview over time. There is also data visualization that displays all of the user's scores and trending performance over time, enabling users to identify areas they have made progress on and where they may need to concentrate on for further improvement. Each user's strengths and weaknesses are identified as a result of multiple evaluations so that users can better understand how they are doing overall. Lastly, the user dashboard compares the user's overall communication skills with technical skills to provide a complete picture of the user's overall performance level. Therefore, the user dashboard provides a valuable resource for encouraging continuous improvement by providing the user with valuable feedback and direction for improvement.

4. Camera, Voice, and Code Editor Modules :

The modules within the system provide a full, multi-modal way to collect responses during the course of an interview. The Camera Module collects facial expressions as well as cues of engagement, allowing the system to assess aspects of non-verbal communication and how engaged or attentive the person being interviewed actually is[3]. The Voice module records audio responses (from the user) and sends them to the system for transcription and evaluation using speech-to-text processing, thereby giving complete and accurate transcription and evaluation of spoken answers. The Code Editor Module is a coding environment located in the browser and uses syntax highlighting as well as code execution and validation of test cases to provide a real-time programming environment with which a user can write, compile, and execute code.

B. Backend Components

The backend intelligence layer is the backbone of the proposed solution and offers the following services: Allows for processing of multimodal inputs, provides a platform for intelligent reasoning; and delivers adaptive iterations and evaluations during an interview.

The backend intelligence layer is built via three components: a speech and text processing service; a dual-centric large language model (LLM) framework; and an analytics and feedback engine. These three components ensure that people are evaluated accurately and provided with guidance that is uniquely tailored to them.

1. Speech and Text Processing Layer :

This layer is responsible for enabling natural and seamless communication between the user and the AI interviewer. It performs speech-to-text conversion to transcribe user audio responses into textual form and text-to-speech synthesis to generate natural-sounding interviewer questions. In addition to transcription and synthesis, the system extracts important voice metadata such as speaking rate, pauses, filler word frequency, pitch variation, and tone. These features are used to analyze communication quality and are forwarded to the evaluation pipeline to support deeper interpretation of user performance.The Speech/Text Processing Layer helps users communicate naturally and effortlessly with the virtual interviewer. This layer converts user audio responses into text and voice imitates text, producing accurate-sounding interviewer questions.

2. AI Processing Layer (Dual-LLM Framework) :

It uses a Dual LLM framework to separate the generation of the questions from the evaluation of the user responses, thus allowing for an unbiased and fair evaluation of the generated questions and user responses. The first model is called the Adaptive Question Generator, which will dynamically generate questions based on the specified role and domain, difficulty level, previous responses by the user, and a history of the user's performance patterns. As the session progresses, the Adaptive Question Generator will continually adjust the complexity of the questions that are presented to the user. For example, if the user is providing a high-quality answer, the Adaptive Question Generator will present difficult, multilayered questions to that user. In contrast, if the user was having difficulty answering questions, the Adaptive Question Generator will either simplify the questions or shift to questions in a related subtopic to help reinforce the user's understanding of the basic concepts.

The second model is the Response Evaluator, which evaluates the user responses to the generated questions by using the transcribed text response as well as the voice and facial recognition data for the user. Response Evaluator will evaluate the user response based on multiple qualitative characteristics such as: relevance, accuracy, depth of description, clarity of description, and level of confidence. By decoupling the generation and evaluation of the questions and user responses, the Dual LLM Framework protects against potential biases that may arise when one model generates questions and evaluates answers, and therefore provides a more reliable and fair way to evaluate user answers.

3. Analytics and Feedback Engine :

The aggregator engine for performance indicators including Technical accuracy Score, Communication Effectiveness Score, Confidence index, and Response Consistency generates metrics from the Evaluator Model's output and creates meaningful performance metrics. It provides real-time micro-feedback as the interview session takes place in the form of guidance for the user to improve concurrently with the Interview (i.e., to improve instantly). After concluding the interview, the feedback engine produces a Post Interview Report summarizing the interviewee's strengths and weaknesses, examples of corrections, and suggested individual improvements. The structure of the post-interview report creates an understanding for the user of what went wrong (i.e., what needs correction) and how to correct it.

C. Database (Secure Storage)

The database layer is developed to give safe, consistent, and privacy-conscious storage of all critical system data. The database contains only user profile data, interview session metadata, and the final evaluation report after each interview session. To protect user privacy and reduce exposure to sensitive data, the system does not keep raw audio and video files in the system as a permanent part. Instead, only feature extractions and summary analytical data are stored in the database for use in providing reports and tracking progress. In addition, all database implementations comply with GDPR regulations and have privacy-conscious management of data, including encryption of data, controls over access to data, and secure authentication policies, providing confidentiality, integrity, and controlled access to user data.

D. Flow of Operation

Once a user establishes an account using Supabase Authentication, they're given a secure session token and are redirected to their personalized dashboard. The user has options for configuring the Interview Parameters using the Interview Orchestration Module via the dashboard. This includes selecting the type of interview (HR, technical, coding, etc.) for various roles/domains, difficulty level, practice mode, and optional time restraints.

After the user has completed their interview configuration in the interview orchestration module, those configurations are sent to the frontend to activate the camera, microphone, and integrated code editor module for the purposes of capturing multimodal responses. The user will provide responses to questions in the interview session through voice, facial expressions (via webcam), Text (only when required), and Coding submissions for Coding Rounds.

Captured audio inputs will be sent to the speech processing layer for converting the audio input to text through speech recognition, and for extracting voice-related metadata such as speaking pace, pauses, and pitch. Simultaneously, each frame of video captured by the camera will be analyzed to detect facial expressions, and engagement by determining the user's level of attention, which will generate facial expression-related metadata. The transcribed text of the user's spoken words and the corresponding voice and facial expression-related metadata will be sent forward to the AI Processing Layer.

The Adaptive Question Generator LLM in this layer generates context-sensitive and adjusted difficulty interviews, creating a dynamic, unique flow for each individual interview session. The Response Evaluator LLM evaluates user responses on a variety of criteria, including how relevant, accurate, deep, clear, and confident the answer is. The response evaluation data is sent to the analytics engine to calculate analytic results and generate real-time, micro-feedback in guided mode to the user throughout their interview experience.

After completing the user's interview, the analytics engine generates a full "final performance" report that contains the user's overall score, identified strengths and weaknesses, and recommendations for future improvement. A record of both an overview of the interview session and the final report are stored securely in the database for use in future reference and to

track progress. The combination of state-of-the-art AI models, multimodal interaction methods, and secure data storage practices will create a sustainable privacy-based interview platform that supports the scalability of the business model for conducting adaptive interview simulations.

VI. Methodology

In the proposed system, we utilize a modular/sequential workflow methodology to enable secure access, multimodal data capture, adaptive AI processing, and comprehensive feedback generation.

User entry through Firebase Authentication provides a secure and personalized login before users can configure their interview session by selecting an interview type (HR or Technical), role, and difficulty level. configuring their interview session. By providing this customization option, users will experience a more realistic, personalized experience during their simulation based on how well they prepared for their interview.

Once sessions have begun, each of the system's modules will be activated to capture multimodal input for the user who is participating in the interview via video and audio recording.

The Camera Module will use the Mediapipe Holistic Model v0.5 to capture the user's facial expressions, body physiology and movements to evaluate their confidence and level of engagement. Similarly, the Voice Module will capture voice audio from the user to convert their verbal responses into text via speech recognition and to analyze their emotional response. For Technical interviews only, users can solve coding problems using an integrated Code Editor Module that supports an online coding environment

Users can code & run programs from their web browser using an application created in React & CodeMirror. When a user selects "Run", the code, language & version selected will be sent to Piston for execution in a secure, sandboxed environment & return standard output, error, and exit codes that will be displayed to the user in real time to provide immediate feedback for their coding performance.

Speech/audio processing utilizes Deepgram's Nova-3 speech-to-text to convert user speech into a text format. Additionally, the interviewer's answers will be displayed in a natural-sounding way back to the user, utilizing Deepgram's Aura-1 text-to-speech, so that users can communicate interactively in real-time with systems.

The AI processing layer consists of a dual LLM architecture. Gemma 3-12B-IT develops adaptive & contextually aware interview questions dynamically on the basis of the user selected domain, role, & previous user responses. 3N-E4B-IT evaluates both text & code response data based upon correctness, clarity, completeness, & relevance. Additionally, it evaluates facial expression, body posture, tone of voice, & emotional state using Mediapipe Holistic & Deepgram tools to evaluate the user's level of confidence & engagement.

The Analytics Feedback Engine provides an overall view of performance by collecting data across several different aspects, such as technical accuracy, communication quality and non-verbal behaviour. While in the interview, the system will provide micro-feedback in real-time, especially while in guided mode. At the end of the interview, a final comprehensive report will be provided outlining strengths and weaknesses, overall scores and actionable suggestions for improvement.

To protect user privacy and compliance, only analytical performance and performance reports are stored in the backend database in Firebase; all raw audio and video files will be deleted once processing of the results has been completed – preserving user privacy while meeting data security standards.

Ultimately, this dashboard provides an opportunity for users to view their history of interviews, coding history, and performance trends over time. Through the use of graphic representation of their skills or abilities, it helps motivate users to assess themselves and formalise learning. Thus, it is an effective method of creating a flexible platform for interview preparation.

VII. Results

An evaluation of the PrepSmartly system assessed its performance in several key areas: how quickly it processes information, how accurately it recognizes faces, and how well it adapts its responses. The tests were performed in a controlled environment using a computer with an Intel i5 processor, 16 GB RAM, a webcam, and a stable internet

connection. The results indicate that PrepSmartly meets the needs for real time interview simulations, providing fast feedback and reliable analysis of user behavior.

A. Prototype Validation and System Functionality

The initial prototype was used for various types of interview (HR, Technical and Coding). It was able to successfully capture and process multi-modal input, including voice responses, face expressions, and coding submissions.

The Interview Interface successfully transcribed voice in real-time, tracked behavior through the camera, and enabled adaptive questioning without impacting any participant visually.

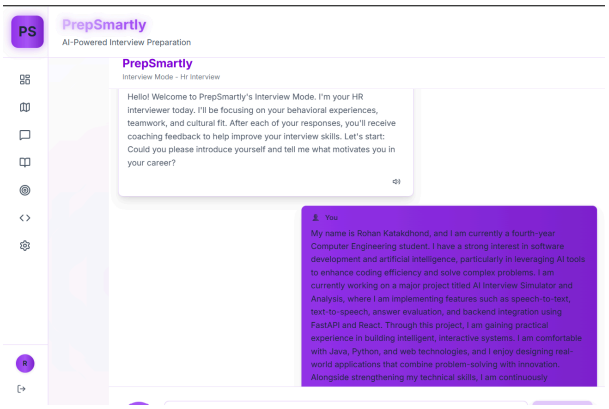

Fig. 2 Real Time Interview Simulation Interface

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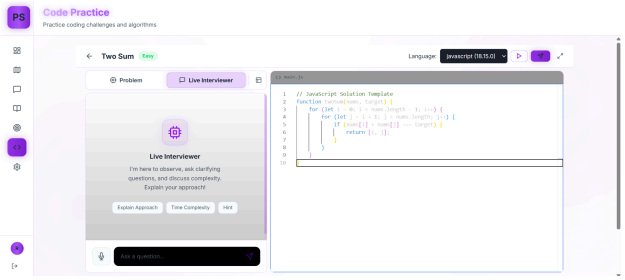

Fig. 3 Code Editor with Real-Time Execution

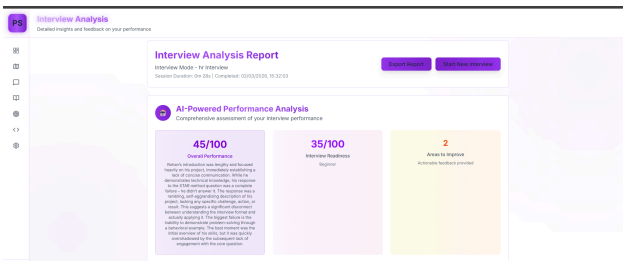

Fig.4-Post-Interview Performance Analytics Dashboard

B. Latency and Real-Time Performance Analysis

System latency was measured as the time between user response submission and AI generated evaluation output.

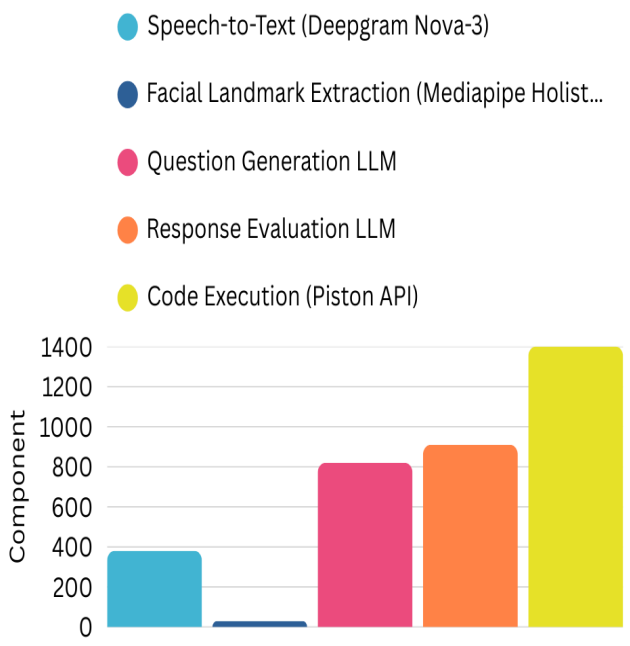

Fig. 5- Average Latency

C. Facial Expression Model Performance

A supervised ML model was made for categorizing face expressions based on the face anatomy as defined using landmarks provided by the MediaPipe Holistic program. The landmarks were used to create coordinate based feature vectors that describe the geometry of face expressions by recording key face regions (eyes, eyebrows, mouth, and nose). The subsequent vectors are used to calculate geometric relationships between the facial anatomy and provide subtle variation in facial muscle behavior across each face expression.

An 80% train/20% test split was used for the dataset. The training portion of the dataset was used to teach the model to translate the facial feature landmarks to five groups that are described by behavioural expressions, including Confident, Neutral, Nervous, Hesitant, and Distracted. The test portion of the dataset was retained for evaluating how well the model is able to generalize on data it has not seen before.

The classifier was shown to perform well on the test data across all behaviours, with respect to head movement and lighting conditions.

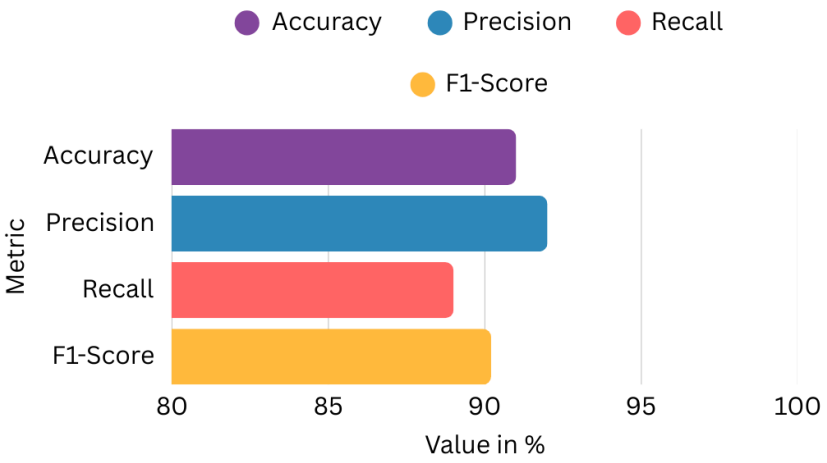

Fig.6-Facial Expression Model Performance Metrics

D. Confusion Matrix Analysis

The simplified matrix focuses on the primary expressive states observed during interview sessions: Confident, Nervous, and Hesitant.

Table.1-Confusion Matrix Analysis of Behavioral Expression Classification

Actual \ Predicted	Confident	Nervous	Hesitant
Confident	112	3	4
Nervous	4	109	6
Hesitant	3	7	102
Neutral	5	7	4

A clear diagonal dominance is shown through the corrected confusion matrix; thus confirming a high level of consistency in classification outcomes across prevalent behaviours. There was the largest degree of classification error between the "Nervous" and "Hesitant" behaviours due to the existence of a significant amount of overlap (in micro-expressions) between these two behaviours; therefore, causing there to be some ambiguity associated with them. An example of this is the similarity of reduced eye contact, some facial tension and a lack of movement of the head to one another when engaged in these two behaviours. Throughout the majority of samples the classifier also demonstrated a high level of consistency within its predictions; thus confirming a high degree of feature learning from facial landmarks.

E. Adaptive LLM Evaluation and Coding Module Performance

The capabilities of the dual LLM were examined through 50 structured mock interviews, evaluating adaptive question behaviour and consistent results. The method adapted the complexity of questions based on previous response(s). When the applicant showed an understanding and provided a structured response, the following questions increased in both their analytical depth and difficulty. If the interviewer experienced difficulties with understanding or incomplete responses, the system then brought forward a simplified follow-up question that reinforced foundation skills. Repeated evaluations, utilising the same responses resulted in a variance of less than 3.2% in grading indicating consistent and fair grading performance.

The test of the coding application was conducted on 60 submissions of programming in three languages, examining each for whether they compiled successfully, were logic-accurate and detected runtime errors.

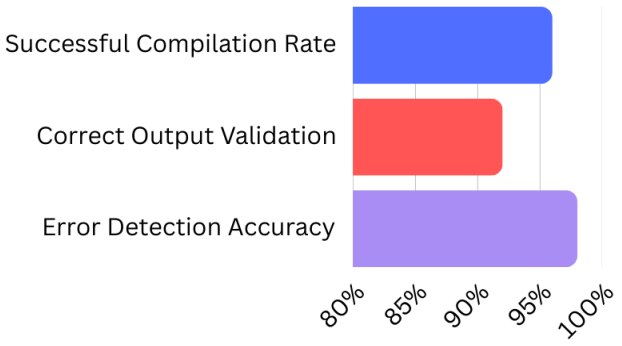

fig.7-Coding Module Performance Metrics Analysis

The PrepSmartly System was able to identify and report on compilation errors, logical mismatches, and run-time errors with the aid of the predefined required structured testing. The execution of the defined responses had an acceptable latency which is necessary to simulate an applicable technical interview..

F. Overall Result Interpretation

The experimental results provide evidence to support that the PrepSmartly system has created a real-time, multimodal interview simulation experience. In addition to low latency times to generate an AI-based response, the system produced reliable facial landmark tracking and successful classification of behavior based on the adaptive LLM framework with reliable and standardized evaluation methods. Furthermore, the coding module was able to execute each program accurately and securely. Overall, these observations confirm that PrepSmartly is a scalable and intelligent method for assessing both technical and behavioral skills as an integrated framework for interview preparation.

VIII. Conclusion

PrepSmartly overcomes the shortcomings of traditional interview preparation by creating an innovative, realistic, adaptive AI-driven multi-mode interview simulation system. Through real-time feedback, dual LLM evaluation and complete performance analytics, this full-service solution provides a comprehensive and customized learning experience. Unlike traditional question-based practice tools, PrepSmartly works by increasing confidence and building communication skills while developing both technical and behavioural abilities.

PrepSmartly utilizes three separate technologies; the Mediapipe Holistic Model v0.5 for behavioural tracking, Deepgram for voice analytics, and AI-based Adaptive Question Generation And Evaluation, to ensure that each candidate is assessed holistically on their technical, verbal and non-verbal skills. The secure architecture used with Appwrite/Firebase allows users to remain anonymous while providing insights into their performance trends and development, among other things.

Overall, PrepSmartly revolutionizes the preparation of interviews as an intelligent, data-driven, and improving experience. It has the potential to provide significant value to all candidates preparing for interviews and increase their likelihood of obtaining employment within today's highly competitive job marketplace.

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